

Jeryn Immanuel S

PROFESSIONAL SUMMARY

Results-driven **AWS Python Developer** with 5+ years of experience with a strong background in developing and deploying scalable serverless applications using AWS services such as **Lambda, S3, SNS, SQS, Cognito, EC2, ECS**, and **Bedrock**. Proficient in leveraging **Pulumi** for infrastructure as code and experienced in implementing document processing pipelines utilizing Amazon **Textract**.

Previously, as a Flutter Developer, designed and developed cross-platform mobile applications with a focus on intuitive UI/UX, integrating RESTful and GraphQL **APIs**, and implementing features like secure user authentication and real-time data synchronization.

Demonstrates excellent communication and analytical skills, with a commitment to continuous learning and delivering high-quality software solutions.

TECHNICAL SKILLS

Programming Language	- Python, Dart
Framework & tools	- Pulumi, Git, FastAPI, Flask, Flutter
Cloud services	- AWS Lambda, S3, SNS, SQS, Cognito, EC2, ECS, Textract, Bedrock
DB	- PostgreSQL, DynamoDB, MongoDB
APIs	- RESTful, GraphQL
CI/CD	- GitHub Actions
Source control	- GitHub, GitLab, and AWS CodeCommit
Error Tracking Tool	- Sentry

EXPERIENCE SUMMARY

- **Web & API Development:**
Extensive experience designing and developing **RESTful** APIs and microservices using

Flask and **FastAPI**, integrating event-driven workflows with AWS Lambda, SQS, and API Gateway. Skilled in database schema design and optimization for **PostgreSQL** and **MongoDB**, ensuring high performance and scalability. Experienced in building backend service layers for enterprise-grade healthcare platforms with secure authentication (OAuth2, JWT) via AWS Cognito.

- **Cloud & DevOps Proficiency:**

Strong understanding of **AWS cloud architecture**, with hands-on experience deploying, monitoring, and scaling applications using **Lambda, ECS (Fargate/EC2), and CloudWatch**. Proficient in **Infrastructure-as-Code (IaC)** using Pulumi, automating resource provisioning for consistent environments. Skilled in CI/CD automation with **GitHub Actions** and AWS CodePipeline to streamline build, test, and deployment workflows.

- **Document Processing & AI/ML:**

Specialized in building automated document processing pipelines leveraging **Amazon Textract** for OCR, table extraction, and key-value pair detection. Experienced in integrating generative AI solutions via Amazon Bedrock for intelligent document classification, summarization, and natural language queries.

- **Automation & ETL:**

Proficient in writing **Python automation scripts** using boto3 to manage AWS resources, orchestrate workflows, and handle data movement between **S3, DynamoDB, and RDS**. Skilled in building ETL pipelines for data transformation and migration across SQL and NoSQL databases.

- **Mobile Application Development:**

Developed cross-platform mobile applications using Flutter and Dart, integrating GraphQL APIs via Hasura for real-time synchronization. Delivered secure, scalable mobile solutions with a focus on user experience, responsive design, and session security.

- **Collaboration & Agile Delivery:**

Proven experience working in **Agile/Scrum** teams, contributing to sprint planning, code reviews, and iterative delivery cycles. Skilled in collaborating with product managers, designers, and QA teams to ensure timely, high-quality releases.

EXPERIENCE

AI-Driven Offboarding Automation Platform - (2025 – Present)

- **Project summary:**

- Leading the design and development of an enterprise-grade SaaS platform that automates large-scale employee offboarding and workforce restructuring. The system integrates with HRIS and payroll systems (e.g., Workday, SAP

SuccessFactors, ADP, BambooHR) to streamline batch terminations, ensure compliance with state/federal laws (e.g., WARN Act, COBRA), and deliver AI-powered insights on cost, compliance, and risk.

- **Duration:** 4 months
- **Roles and Responsibilities:**
 - Led a cross-functional engineering team through end-to-end development, from technical design to production deployment.
 - Collaborated with product and HR domain experts to translate complex offboarding and compliance workflows into scalable, maintainable APIs.
 - Designed backend APIs, data models, and integration pipelines supporting batch offboarding, document generation, and analytics.
 - Directed the AI integration strategy, identifying use cases for predictive analytics, document intelligence, and risk scoring.
 - Oversaw development best practices, conducted code reviews, and managed Agile sprint cycles with regular stakeholder updates.
 - Built integrations with major HRIS systems for real-time data synchronization and post-offboarding updates.
 - Implemented secure authentication (SSO, OAuth2, MFA) and Role-Based Access Control (RBAC) for enterprise security.
- **Key Technical Contributions & Achievements:**
 - Architected scalable backend services for batch employee processing (100+ employees per run), integrating HRIS APIs for data ingestion and updates.
 - Designed and led implementation of AI-powered modules for:
 - ♣ **Employee Identification:** ML-driven scoring to recommend offboarding candidates based on performance, tenure, and cost.
 - ♣ **Adverse & Financial Impact Reporting:** Predictive analytics to forecast severance costs and highlight compliance or bias risks.
 - ♣ **Document Generation:** AI-assisted template management with compliance validation and dynamic placeholders (e.g., WARN, COBRA).
 - Integrated AI using Thunai AI for document generation, natural language queries, and risk prediction.
 - Designed microservice architecture with FastAPI and asynchronous communication via SQS/SNS for workflow orchestration.
 - Configured CloudWatch monitoring, auditing, and billing modules for operational transparency and reliability.
 - Guided optimization of database queries, caching, and scaling strategies to improve system performance and cost efficiency.
 - Troubleshooted production issues and optimized distributed systems to maintain high availability with minimal downtime.

- **Technologies/Tools:** Python, FastAPI, Flask, PostgreSQL, AWS (Lambda, API Gateway, S3, CloudWatch, Cognito, SQS, SNS), HRIS APIs (Workday, BambooHR, ADP), Pulumi, GitHub Actions, Docker, ECS (Fargate).

Prior Authorization Automation Platform - AWS Python (2023 – 2025)

- **Project summary:**
 - Backend API development for an enterprise-grade healthcare platform that automates prior authorization (PA) and integrates with Electronic Health Records (EHRs), optimizing workflows and ensuring CMS compliance.
- **Duration:** 2 years 8 months
- **Roles and Responsibilities:**
 - Gathered and analyzed requirements to define technical scope and design objectives.
 - Designed backend API web services and data schemas tailored for scalable, healthcare-specific workflows.
 - Deployed Lambda-based backend services and exposed them through API Gateway for secure, scalable access.
 - Proactively monitored logs, metrics, and alerts to ensure system reliability and performance.
 - Collaborated with cross-functional teams in Agile/Scrum environments, participating in sprint planning, code reviews, and daily standups.
- **Key Technical Contributions & Achievements:**
 - Architected and deployed **serverless applications** using AWS Lambda, integrating with S3, SNS, SQS, DynamoDB, and Cognito for scalable, event-driven systems.
 - Implemented **Infrastructure as Code (IaC)** using Pulumi to automate provisioning of AWS resources (Lambda, S3, API Gateway, IAM roles, ECS clusters).
 - Built **document processing pipelines** leveraging Amazon Textract for OCR, key-value extraction, and table detection, improving efficiency for large-scale workflows.
 - Integrated **Amazon Bedrock** to deliver AI-powered document classification, summarization, and natural language query handling.
 - Designed and managed **containerized applications** using Amazon ECS (Fargate) and EC2 with a focus on high availability and cost optimization.
 - Created and optimized **CI/CD pipelines** with GitHub Actions and AWS CodePipeline, reducing release cycles and improving reliability.
 - Developed **Python automation scripts** using boto3 for AWS resource management, workflow triggering, and data transfers.

- o Built **REST APIs** and event-driven integrations with API Gateway, Lambda, and SQS to decouple systems and improve fault tolerance.
 - o Implemented **secure authentication** with AWS Cognito, including OAuth2 and JWT-based session management.
 - o Configured **monitoring and alerting** with Amazon CloudWatch, using alarms and custom metrics for proactive system health checks.
 - o Conducted **performance tuning and cost optimization**, including right-sizing of AWS resources to reduce operational costs.
 - o Troubleshoot production issues and optimized distributed systems to maintain high availability with minimal downtime.
- **Technologies/Tools:** Python, Flask, FastAPI, Postgres, AWS - Lambda, API gateway, Textract, S3, Cloudwatch, GitHub Actions

Barberaa – Admin Platform Development (2022)

- **Project summary**
 - o Barberaa is a salon and barber discovery platform where users can find nearby barbers and salons. I worked on the **admin side of the platform**, enabling barbers and salon owners to register their store or franchise, manage their profiles, and update service offerings.
- **Duration:** 1 years
- **Roles & Responsibilities:**
 - o Designed and developed **REST APIs** in Flask to handle registration, profile management, and data retrieval for salon owners.
 - o Integrated APIs with **MongoDB** for data storage and retrieval, ensuring accuracy and reliability.
 - o Collaborated with frontend developers to connect APIs with the UI and support a smooth user experience.
 - o Participated in **Agile ceremonies** using Jira for sprint planning, backlog refinement, and tracking deliverables.
- **Key Technical Contributions & Achievements:**
 - o Built and optimized the entire **backend service layer**, reducing onboarding time for new barbers and franchises.
 - o Implemented **database query optimizations** in MongoDB, improving response times and overall platform performance.
 - o Utilized **Docker** for containerization, ensuring consistency across development, staging, and production environments.
 - o Developed **Jinja-based admin templates** for quick internal management of barber and salon data.

- Contributed to **team-based development workflows**, performing code reviews and supporting collaborative debugging sessions.
- **Technologies/Tools:** Flask, MongoDB, Docker, REST APIs, Jinja, Jira, Git

Self-service Mobile Application (2021)

- **Project summary**
 - Designed and developed a cross-platform mobile application using Flutter to provide a seamless user experience across Android and iOS. The app supported real-time data synchronization, secure authentication, and polished UI features to improve user engagement and operational efficiency.
- **Duration:** 1 year
- **Roles & Responsibilities:**
 - Designed and developed cross-platform mobile applications using Flutter and Dart.
 - Collaborated with product managers, designers, and backend engineers in Agile sprints to ensure timely delivery.
 - Translated Zeplin UI/UX designs into responsive and interactive application interfaces.
 - Implemented secure user authentication and session management using encrypted local storage.
 - Conducted unit and integration testing to ensure reliability and scalability of the application.
- **Key Technical Contributions & Achievements:**
 - Integrated **GraphQL APIs** via Hasura for real-time data synchronization, resulting in improved responsiveness and user engagement.
 - Built **custom UI components and interactive features** aligned with design specifications, creating a polished, intuitive user experience.
 - Employed **secure storage solutions** to meet compliance requirements and protect sensitive user data.
 - Utilized **Git** for version control and collaborative development, enabling smooth code reviews and continuous integration.
 - Performed **testing and release management** to guarantee smooth production rollouts with minimal post-release issues.
- **Technologies/Tools:** Flutter, Dart, Hasura GraphQL, Secure Storage, Zeplin, Git, Postman

EDUCATION

2012 – 2016	Karunya Institute of Science and Technologies, Coimbatore	• Bachelor of Technology • 6.81 CGPA
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DECLARATION

I hereby declare that all the details mentioned above are true to the best of my knowledge and belief.

Place: Nazareth

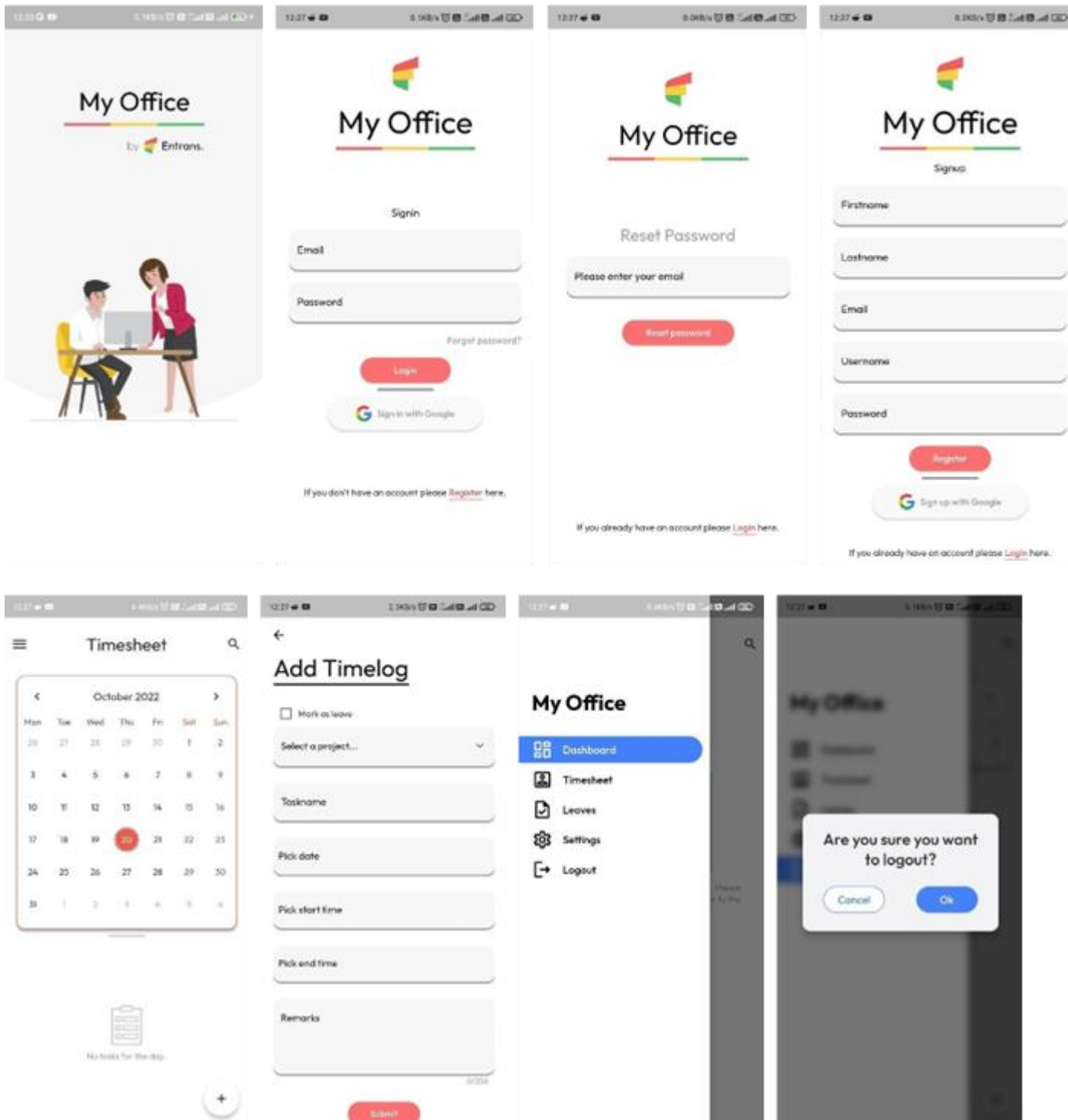
Jeryn Immanuel S

CONTACT INFORMATION

Mobile no 7010555507

E-mail jeryn123@gmail.com

Screenshots – My-Office



Screenshots – Self Service

